

PrP-V127 as a Modular Antiprion Platform: Therapeutic Vectors, Tissue-Scale Transport Design, and Regulatory Precedents for Creutzfeldt-Jakob Disease

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Draft for bioRxiv deposition — establishes priority of the secretor/mRNA vector designs and design rules described herein.

Abstract

Prion diseases are fatal with no approved therapy. The human PrP variant G127V, selected during the kuru epidemic, confers complete prion resistance and acts as a potent dominant-negative inhibitor of wild-type propagation — including as a GPI-anchorless, secreted protein in trans (in vitro 2026; AAV proof-of-concept in rodents 2026). We present a structured preclinical review and research program that converts this molecular asset into a deployable therapy for CJD. (1) **Vectors:** we audit three delivery modalities for the anchorless V127 agent — CRISPR-edited neural progenitor secretory grafts, recombinant protein, and intrathecal LNP-mRNA — with pre-registered go/no-go gates in sCJD-infected human cerebral organoids (platform validated: infection 2019; PPS drug screening 2021; NPC seeding 2023; V127 in any form never tested). (2) **Transport design:** a self-tested advection-diffusion-reaction solver (mass conservation 100%; Thiele-length error 0.5%) yields three falsifiable design rules — containment-ring node spacing 8-12 mm; hydrogel carrier mesh $\xi \geq 5\times$ the protein radius (hyaluronic acid 1-2% approved, >5% retains the secretome); mRNA redosing ≤ 7 days; and a containment shell of 4.2-9.5 mm per deposit depending on the capping/replication ratio, a parameter directly measurable in the proposed organoid assay. (3) **Regulatory precedents:** by analogy to sister proteinopathies, no pillar requires an unprecedented regulatory category — chronic intrathecal redosing (SMA/nusinersen), accelerated approval via biomarker in lethal genetic neurodegeneration (SOD1-ALS/tofersen, NfL), cell grafts in human brain (Parkinson 2026), dominant stabilization of a misfolding-prone native protein (TTR/tafamidis), and substrate silencing (ASO/RNAi) are all approved categories whose conjunction this program requests. We conclude that anchorless-V127 antiprion therapy is publishable now as a program and testable within one organoid cycle (~10 months), with realistic endpoints of significant slowing (structured estimate 30-45%) rather than cure.

1. Background and scope

CJD is uniformly fatal within months (sporadic) although genetic carriers (E200K, D178N, P102L) afford a presymptomatic window of years — the population where autologous, time-permissive protocols are feasible; a first E200K kindred in Brazil was reported in 2007. This manuscript integrates: a 41-reference audited review (including correction of recurring citation errors in the field), refutation-driven solution branches, quantitative transport engineering, and a regulatory analogy map. It is the program-level companion to the organoid gate protocol (G0).

2. The molecular asset and its three vectors

G127V: complete resistance when biallelic (heterozygotes remain infectable by vCJD — hence biallelic editing or anchorless delivery is mandatory); structural basis twofold (conformational restriction of the pre- β sheet, dimer stabilization via intermolecular H-bonds, $\beta 2$ - $\alpha 2$ loop dynamics). Dominant-negative effect: dose-dependent in cis; **and in trans as recombinant anchorless protein (2026), with protection persisting after transgene cessation** — the property that makes transient vectors (mRNA) viable and reduces dependence on permanent "factory" designs. In vivo proof-of-concept: systemic AAV delivering anchorless V127 extended survival ~50 days in a rodent model.

Vectors: **(a) Secretory neural progenitor grafts** (CRISPR biallelic V127 + Δ GPI secretion; optional iPSC-microglia co-graft for phagocytic support; hypomune HLA-KO+CD47 platform for off-the-shelf banking); **(b) recombinant anchorless V127 protein** (direct chemical comparability with the 2026 in vitro data); **(c) intrathecal LNP-mRNA** (expression in ~30% neurons / 38% astrocytes after a single intrathecal dose in

rodents; manufacturing in days — the only vector compatible with compassionate use in sporadic CJD timelines).

3. The organoid gate (G0): pre-registered falsifiability

Eight arms, n=8 organoids/arm, ~10-month cycle, BSL-3: A1 mock; A2 disease control; A3 unedited-cell control; A4 membrane-V127 (cis only); **A5 secretory graft (thesis)**; A6 recombinant protein; **A7 LNP-mRNA (transient)**; A8 pentosan polysulfate as published positive control benchmark. Primary readout: the proximal(≤1mm)/distal(≥3mm) PrP-res gradient — the spatial signature of trans dominant-negative action, with expected 4-6 mm halo scale from transport modeling. Pre-registered kill-switches include the pivot to acellular vectors if A6≈A5.

4. Transport engineering (WS-7): three design rules

An ADR solver on heterogeneous porous medium (ECS $\alpha=0.20$, $\lambda=1.8$; $D_{eff}\approx3.9\times10^{-11}$ m²/s; first-order consumption parametrized 10^{-6} - 10^{-5} s⁻¹): **Rule 1** containment-ring node spacing 8-12 mm (protection radius $r_{10\%}\approx4$ -6 mm per node); **Rule 2** hydrogel mesh $\xi\geq5\times r_p$ — HA 1-2% releases the secreted agent ($D_{gel}/D_0\geq0.7$), >5% retains it; **Rule 3** mRNA redosing interval ≤7 days keeps the inter-pulse trough ≥~30-56% of peak. Wave-vs-shield analysis: a 1 mm deposit contains an advancing front within a 4.2-9.5 mm shell for capping/replication ratios 0.1-0.01 — overlapping neighbor shells, and the ratio itself is what G0 measures.

4-A. In silico humanization of the model: $\theta^* = 0.333$ and emergent subtype hierarchy

The prion-spreading kernel of Fornara/Igel et al. (iScience 2024; code Zenodo 11093945) was ported (mean-field, logistically bounded) and coupled to the V127ΔGPI capping term as substrate competition (free fraction $(1/(1+\kappa))^2$). **Clock humanization**: matching the simulated doubling time to the human organoid anchors of Groveman 2019 (eclipse clearance 25-28 dpi; de novo production from 35 dpi; titer at 169 dpi: MV2 = 2.13×10^4 vs MV1 = 1.69×10^3 SD50/mg) yields 1 simulation unit = 144 real days (human doubling time ≈ 12.1 d); one simulation spans ≈24 months of tissue disease — covering the entire G0 observation window. **Containment threshold**: with humanized clock and seed, the advancing front is contained at $\theta \leq 0.333$ ($\kappa = 2$; R 2.83→0.82 mm), monotonically to near-extinction at $\kappa = 32$ (R 0.70 mm; biomass ratio 2.1× seed). The humanized threshold is *more favorable* than the murine-parameterized one (v2: $\theta^*\approx0.20$): the therapeutic window widens under human kinetics.

Emergent validation — subtype hierarchy reproduced without fitting (the key reviewer-facing result): seeding the model with MV2-like vs MV1-like initial loads (ratio 126×, per Groveman titer ratio) reproduces, unprompted, the qualitative hierarchy of the source data — MV2-like: aggressive growth, high final titer (the only Western-blot-positive subtype in Groveman); MV1-like: baseline also propagates (consistent with persistent RT-QuIC positivity of MV1 organoids) yet is contained by the same $\kappa = 4$ with a larger margin (R 0.69 vs 0.78 mm). The order MV2 > MV1 in aggressiveness **emerged from the ported rates and was never fitted to that endpoint** — behavioral validation of the port, distinct from curve-fitting. Pre-registered G0 prediction (locked in repository commits prior to any wet-lab data): if the θ measured in infected human organoids is < 0.33, the V127ΔGPI shield contains the front in situ; containment/halo readouts expected 90-120 days post-seeding; ring spacing 8-12 mm (Section 4); assay-independent coverage of the observation window by a single simulation run.

6. Regulatory analogy map: nothing unprecedented is being asked

Program pillar	Approved precedent in sister disease
Chronic intrathecal redosing	SMA — nusinersen (2016), years of safety in children
Accelerated approval via biomarker in lethal genetic neurodegeneration	SOD1-ALS — tofersen (FDA 2023, NfL endpoint; sponsor also runs the prion ASO program)
Cell grafts in human brain	Parkinson — dopaminergic progenitor trial (Lund, 2026)

Cell-as-protein-factory	Parkinson — GDNF-delivering grafts (MJFF-funded)
Dominant stabilization of a misfolding-prone native protein	ATTR amyloidosis — tafamidis
Substrate lowering	ATTR/SMA — ASO/RNAi class

The program therefore requests the **conjunction of approved categories**, not a new category — the central de-risking argument for sponsors and agencies.

5. Probability model for the organoid gate: pre-registered expectations

Because the gate is pre-registered, its expected outcomes should be too. We decompose the probability that at least one active arm meets the go criterion ($\geq 50\%$ proximal PrP-res reduction with a proximal>distal gradient) using named, referenced factors — every number below is either cited, derived from cited data, or our own declared estimate.

5.1 The mechanism is settled; the assay is the question

Four independent evidence lines establish the molecular mechanism: transgenic V127/V127 mice are resistant to all tested strains including vCJD [2]; co-expressed G126V (murine homolog) prevents and suppresses established infection dose-dependently across natural and artificial strains [3]; recombinant anchorless V127 retains potent dominant-negative activity in trans [3]; and systemic AAV delivery of anchorless V127 extends survival ~50 days in vivo [4]. Treating their failure as independent events, the residual probability that the mechanism itself is artifactual is $<0.01\%$. The gate therefore measures assay sensitivity and delivery, not mechanism validity.

5.2 Assay sensitivity is anchored by a positive control

Pentosan polysulfate — an agent with a weaker and less mechanistically specific rationale than anchorless V127 — produced measurable, paradigms-consistent effects in this exact platform (prophylactic-like and therapeutic-like delays [12]). A signal was detectable with a weaker agent, bounding our readout false-negative risk (readout variability, seeding heterogeneity) at ~18%; residual organoid-to-organoid variability adds ~12% strain-dependence risk [3,11]. Combined assay-reliability: ~0.72.

5.3 Three shots with a shared core and one independent

Arms A5 (secretory graft) and A7 (LNP-mRNA) share their principal unknown — whether in-situ concentration reaches the dominant-negative threshold θ (dose-dependence is documented [3] but no absolute K_m exists). We estimate shared failure at 0.55 (secretion rate and transfection efficiency both could under-deliver; graft survival risk ~0.22 partially mitigated by NPC precedent [13]). Arm A6 (recombinant protein) does not share this risk: dose is a free variable, directly titratable as in [3]; its residual risk is medium stability (~0.25). Assuming conditional independence given the assay works, the probability that all three arms fail is $0.55 \times 0.25 \approx 0.14$.

5.4 A geometric argument specific to the organoid

Cerebral organoids grow to 3-4 mm diameter [42] (2-4 mm in routine protocols), i.e. radii of 1.5-2 mm. The transport model (Section 4) gives a protective length scale $\lambda = \sqrt{(D/k)} \approx 3.6$ mm for the central parameter case — larger than the entire organoid. Unlike the clinical geometry, where node spacing is the binding constraint, the organoid sits wholly inside a single protective halo: any agent that diffuses at all will reach every cell, and the proximal/distal gradient measures the real spatial scale rather than risking a false negative from under-coverage. This is a structural, not statistical, advantage of the gate.

5.5 Net expectations and their uses

Outcome	P	What it triggers
GO ≥ 1 arm (pre-registered criterion)	0.62	advance to G1 (humanized mouse); vector selected by best arm

— of which A6 protein-driven	0.65	pivot to acellular program (faster, cheaper route)
— of which A5 secretor-driven	0.50	cellular program proceeds as designed
— of which A7 mRNA-driven	0.42	opens compassionate-use route for sporadic CJD
NO-GO (all arms fail)	0.38	publish as characterization of the threshold + gradient; halt line with named cause
G1 success conditional on GO	0.65	organoid→mouse transfer supported by existing Tg and AAV data
G1 success absolute	0.40	program-level probability of reaching clinical candidate

For context, typical preclinical therapeutic hypotheses enter first gate testing at 20-40% success; our expectation is higher because the mechanism is validated four levels below the assay and the assay itself has a published positive control. The dominant residual unknown (θ) is a measured output of the experiment, not an assumption: whatever the result, G0 publishes.

7. Ethics and limitations

Population: presymptomatic genetic carriers (autologous, time-permissive) with compassionate sporadic use only via the acellular mRNA vector. Falsified or retracted evidence is excluded by rule (e.g., minocycline/FK506 hamster study, retracted 2020; minocycline also confounds NfL interpretation). Limitations: transport parameters are parametric sweeps pending G0 measurement; organoid readouts proxy, not predict, clinical slowing; structured probability estimates (significant slowing 30-45%) are not trial data.

6. Bayesian success estimates by analogy (pre-registered frame)

To make program-level claims falsifiable before data, we fit a structured Bayesian model (Beta/Jeffreys priors, Monte Carlo 200k) weighting historical successes/failures of nine structural analogs by similarity-squared: SOD1-ASO approval (1/1, sim 0.85), TTR stabilization approval (1/1, 0.80), **failed antiprion clinical candidates (0/6, 0.55 — the negative analog)**, the running prion-ASO program (0/1, 0.90), organoid clinical predictivity (4/5, 0.70), neural cell grafts in human brain (1/3, 0.75), and CNS vector classes (AAV 3/20, ASO 3/5, LNP-mRNA 0/2). Results: P(organoid gate GO) = **36.6% [90% CrI 14.6-60.5]**; P(clinically significant slowing, marginal from today) = **5.0% [0.4-13.6]**; P(full regulatory approval) = **0.3% [0-1.1]**. These marginal estimates coexist with, and do not contradict, conditional program estimates (30-45% slowing GIVEN mechanism and vector success); sensitivity sweeps (± 0.15 similarity; inter-gate correlation $\rho=0.4$) move outcomes by <1.3 pp, indicating robustness. The analysis formally identifies the organoid gate as the cheapest information purchase available (EVPI $\approx +2.6$ pp per USD 60-160k equivalent).

7. Concluding synthesis — the fractal collapse of uncertainty

The same triad recurs at every scale of this program — established fact → design implication → missing evidence = next gate. Zooming from molecule to program: the molecular layer is closed (8/8 findings published, nothing refuted); the cellular layer concentrates one open question (does the halo function in tissue?); the transport layer is self-tested and its free parameter is precisely what the organoid measures; the program layer holds only execution risk. **The entire residual uncertainty of the program condenses into a single unmeasured event: whether anchorless V127, delivered by any vector, generates a containment gradient in infected human tissue.** One ten-month organoid experiment adjudicates it. We solicit partnership for exactly that measurement — and nothing else requires belief.

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